

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** Cobalt powder (Irregular spherical shape)  
**Cat No. :** S60348  
**CAS No** 7440-48-4  
**Molecular Formula** Co

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**
**Importer**

Fisher Scientific Korea  
 D5,D6, Incheon Airport Logistics Complex  
 150, Gonghangdong-Ro 296 Beon-Gil  
 Jung-Gu, Incheon  
 Tel: +82-1661-9555  
 Fax: +82-2-2023-0603

**Supplier**

Alfa Aesar  
 Thermo Fisher Scientific Chemicals, Inc.  
 30 Bond Street  
 Ward Hill, MA 01835-8099

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

**Emergency telephone:** Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

Flammable solids Category 2

**Health hazards**

|                                             |            |
|---------------------------------------------|------------|
| Acute Oral Toxicity                         | Category 4 |
| Acute Inhalation Toxicity - Dusts and Mists | Category 1 |
| Serious Eye Damage/Eye Irritation           | Category 2 |
| Respiratory Sensitization                   | Category 1 |
| Skin Sensitization                          | Category 1 |
| Germ Cell Mutagenicity                      | Category 2 |
| Carcinogenicity                             | Category 1 |
| Reproductive Toxicity                       | Category 1 |

**Environmental hazards**

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Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 1

## Label Elements



Signal Word

Danger

### Hazard Statements

H228 - Flammable solid  
H302 - Harmful if swallowed  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H330 - Fatal if inhaled  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects if inhaled  
H360 - May damage fertility or the unborn child  
H350i - May cause cancer by inhalation  
H410 - Very toxic to aquatic life with long lasting effects  
H400 - Very toxic to aquatic life

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P264 - Wash hands and face thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P271 - Use only outdoors or in a well-ventilated area  
P284 - Wear respiratory protection  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P273 - Avoid release to the environment

#### Response

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction  
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor  
P320 - Specific treatment is urgent (see supplemental first aid instructions on this label)  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P321 - Specific treatment (see supplemental first aid instructions on this label)  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P337 + P313 - If eye irritation persists: Get medical advice/attention

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P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P391 - Collect spillage

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
4

Flammability  
2

Instability  
0

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component      | Common Name              | CAS No    | Index No | Weight % |
|----------------|--------------------------|-----------|----------|----------|
| Cobalt, powder | No information available | 7440-48-4 | KE-06060 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

None under normal use conditions.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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## Control Parameters

| Component      | CAS No    | Korea                       | ACGIH TLV                   | OSHA PEL                                                            |
|----------------|-----------|-----------------------------|-----------------------------|---------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | TWA: 0.02 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> | (Vacated) TWA: 0.05 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |

| Component      | CAS No    | European Union | The United Kingdom                                                                   | Germany |
|----------------|-----------|----------------|--------------------------------------------------------------------------------------|---------|
| Cobalt, powder | 7440-48-4 | Not listed     | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. | Haut    |

## ACGIH - Biological Exposure Indices

| Component      | CAS No    | ACGIH - Biological Exposure Indices                                                      |
|----------------|-----------|------------------------------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | 15 µg/L<br>Medium: urine<br>Time: end of shift at end of workweek<br>Determinant: Cobalt |

## Exposure Controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                      |                                                                                                                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal protective equipment</b> | Use only those certified by the Korea Occupational Safety and Health Administration.                                                                                                                                          |
| <b>Respiratory Protection</b>        | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                               |
| <b>Recommended Filter type:</b>      | Particulates filter conforming to EN 143<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

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|                                          |                          |                                   |
|------------------------------------------|--------------------------|-----------------------------------|
| Appearance (Physical State, Color, etc.) | Solid Powder             |                                   |
| Odor                                     | No information available |                                   |
| Odor Threshold                           | No data available        |                                   |
| pH                                       | No information available |                                   |
| Melting Point/Range                      | 1495 °C / 2723 °F        |                                   |
| Softening Point                          | No data available        |                                   |
| Boiling Point/Range                      | 2870 °C / 5198 °F        |                                   |
| Flash Point                              | No information available | Method - No information available |
| Evaporation Rate                         | Not applicable           | Solid                             |
| Flammability (solid,gas)                 | No information available |                                   |
| Explosion Limits                         | No data available        |                                   |
| Vapor Pressure                           | No data available        |                                   |
| Vapor Density                            | Not applicable           | Solid                             |
| Specific Gravity / Density               | No data available        |                                   |
| Bulk Density                             | No data available        |                                   |
| Water Solubility                         | Insoluble                |                                   |
| Solubility in other solvents             | No information available |                                   |

## Partition Coefficient (n-octanol/water)

| Component      | CAS No    | log Pow           |
|----------------|-----------|-------------------|
| Cobalt, powder | 7440-48-4 | No data available |

|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | No data available        |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |
| Molecular Formula         | Co                       |       |
| Molecular Weight          | 58.93                    |       |

## SECTION 10: STABILITY AND REACTIVITY

|                   |                                            |
|-------------------|--------------------------------------------|
| <u>Reactivity</u> | None known, based on information available |
|-------------------|--------------------------------------------|

|                           |                                 |
|---------------------------|---------------------------------|
| <u>Chemical Stability</u> | Stable under normal conditions. |
|---------------------------|---------------------------------|

|                                           |                               |
|-------------------------------------------|-------------------------------|
| <u>Possibility of Hazardous Reactions</u> |                               |
| Hazardous Polymerization                  | No information available.     |
| Hazardous Reactions                       | None under normal processing. |

|                            |             |
|----------------------------|-------------|
| <u>Conditions to Avoid</u> | None known. |
|----------------------------|-------------|

## Incompatible Materials

ALFAAS60348

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None known.

## Hazardous Decomposition Products

None under normal use conditions.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

**Inhalation** May produce an allergic reaction. Very toxic by inhalation. Avoid breathing dust or spray mist.  
**Ingestion** May cause allergic reaction. May be harmful if swallowed.  
**Eyes** Avoid contact with eyes. Irritating to eyes. Sensitization.  
**Skin** Avoid contact with skin. May cause irritation. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons.

#### Information on Health Hazards

##### (a) acute toxicity;

**Oral** Category 4  
**Dermal** No data available  
**Inhalation** Category 1

| Component      | CAS No    | LD50 Oral                 | LD50 Dermal       | LC50 Inhalation     |
|----------------|-----------|---------------------------|-------------------|---------------------|
| Cobalt, powder | 7440-48-4 | LD50 = 6171 mg/kg ( Rat ) | No data available | 10 mg/L ( Rat ) 1 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

**Respiratory** Category 1  
**Skin** Category 1

| Component      | CAS No    | Test method       | Test species      | Study result      |
|----------------|-----------|-------------------|-------------------|-------------------|
| Cobalt, powder | 7440-48-4 | No data available | No data available | No data available |

No information available

(e) germ cell mutagenicity; Category 2

| Component      | CAS No    | Test method       | Test species      | Study result      |
|----------------|-----------|-------------------|-------------------|-------------------|
| Cobalt, powder | 7440-48-4 | No data available | No data available | No data available |

(f) carcinogenicity; Category 1

| Component      | CAS No    | Test method       | Test species / Duration | Study result      |
|----------------|-----------|-------------------|-------------------------|-------------------|
| Cobalt, powder | 7440-48-4 | No data available | No data available       | No data available |

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | CAS No | IARC | NTP | ACGIH | OSHA | UK |
|-----------|--------|------|-----|-------|------|----|
|-----------|--------|------|-----|-------|------|----|

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|                                                                    |           |                                                                          |                        |    |   |            |
|--------------------------------------------------------------------|-----------|--------------------------------------------------------------------------|------------------------|----|---|------------|
| Cobalt, powder                                                     | 7440-48-4 | Group 2A                                                                 | Reasonably Anticipated | A3 | X | Not listed |
| IARC (International Agency for Research on Cancer)                 |           | IARC (International Agency for Research on Cancer)                       |                        |    |   |            |
|                                                                    |           | Group 1 - Carcinogenic to Humans                                         |                        |    |   |            |
|                                                                    |           | Group 2A - Probably Carcinogenic to Humans                               |                        |    |   |            |
|                                                                    |           | Group 2B - Possibly Carcinogenic to Humans                               |                        |    |   |            |
| NTP: (National Toxicity Program)                                   |           | NTP: (National Toxicity Program)                                         |                        |    |   |            |
|                                                                    |           | Known - Known Carcinogen                                                 |                        |    |   |            |
|                                                                    |           | Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen |                        |    |   |            |
| ACGIH: (American Conference of Governmental Industrial Hygienists) |           | A1 - Known Human Carcinogen                                              |                        |    |   |            |
|                                                                    |           | A2 - Suspected Human Carcinogen                                          |                        |    |   |            |
|                                                                    |           | A3 - Animal Carcinogen                                                   |                        |    |   |            |
|                                                                    |           | ACGIH: (American Conference of Governmental Industrial Hygienists)       |                        |    |   |            |

(g) reproductive toxicity; Category 1

| Component      | CAS No    | Test method       | Test species / Duration | Study result      |
|----------------|-----------|-------------------|-------------------------|-------------------|
| Cobalt, powder | 7440-48-4 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

## Other Adverse Effects

Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

| Component      | CAS No    | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|----------------|-----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component      | CAS No    | Freshwater Fish                                  | Water Flea        | Freshwater Algae  | Microtox          |
|----------------|-----------|--------------------------------------------------|-------------------|-------------------|-------------------|
| Cobalt, powder | 7440-48-4 | LC50: > 100 mg/L, 96h static (Brachydanio rerio) | No data available | No data available | No data available |

## Persistence and degradability

**Persistence** Insoluble in water.  
**Degradability** Not relevant for inorganic substances.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential** May have some potential to bioaccumulate

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## Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility.

## Ozone Depletion Potential

| Component      | CAS No    | Ozone Depletion Potential |
|----------------|-----------|---------------------------|
| Cobalt, powder | 7440-48-4 | Not listed                |

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN3089                          |
| Proper Shipping Name    | METAL POWDER, FLAMMABLE, N.O.S. |
| Technical Shipping Name | Cobalt, powder                  |
| Hazard Class            | 4.1                             |
| Packing Group           | II                              |

### IATA

|                         |                                 |
|-------------------------|---------------------------------|
| UN-No                   | UN3089                          |
| Proper Shipping Name    | METAL POWDER, FLAMMABLE, N.O.S. |
| Technical Shipping Name | Cobalt, powder                  |
| Hazard Class            | 4.1                             |
| Packing Group           | II                              |

### IMDG/IMO

|                         |                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| UN-No                   | UN3089                                                                                                   |
| Proper Shipping Name    | METAL POWDER, FLAMMABLE, N.O.S.                                                                          |
| Technical Shipping Name | Cobalt, powder                                                                                           |
| Hazard Class            | 4.1                                                                                                      |
| Packing Group           | II                                                                                                       |
| Marine Pollutant        | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |

## **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

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## International Inventories

| Component      | CAS No    | KECL     | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|----------------|-----------|----------|------|-----------|-------|-----|------|-------|------|------|------|
| Cobalt, powder | 7440-48-4 | KE-06060 | X    | 231-158-0 | X     | X   | -    | X     | X    |      | X    |

| Component      | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component      | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|----------------|-----------|----------|------------------------------|---------------------------|
| Cobalt, powder | 7440-48-4 | Listed   | Not applicable               | Not applicable            |

## Korean National Regulations

| Component      | CAS No    | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|----------------|-----------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Cobalt, powder | 7440-48-4 | Annex 1 - KE-06060                                                  | Not applicable       | Not applicable                              |

| Component      | CAS No    | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|----------------|-----------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component      | CAS No    | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|----------------|-----------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component      | CAS No    | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|----------------|-----------|-------------------|------------------------------------|-----------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable    | Listed                             | CMR                                                       |

| Component      | CAS No    | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|----------------|-----------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Cobalt, powder | 7440-48-4 | Listed                                                       | Not applicable               | Not applicable                         |

| Component      | CAS No    | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|----------------|-----------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Cobalt, powder | 7440-48-4 | Listed                               | Listed                                             | 0.02 mg/m <sup>3</sup> TWA         |

| Component      | CAS No    | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|----------------|-----------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                                              | TWA: 0.02 mg/m <sup>3</sup>                    | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component | CAS No | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-----------|--------|----------------------------|---------------------------|------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
|           |        |                            |                           |                                                |                             |                                    |                             |

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|                |           |                |                |                                         |                |                |                |
|----------------|-----------|----------------|----------------|-----------------------------------------|----------------|----------------|----------------|
|                |           |                |                | and Dangerous<br>Substances<br>When Wet |                |                |                |
| Cobalt, powder | 7440-48-4 | Not applicable | Not applicable | Not applicable                          | Not applicable | Not applicable | Not applicable |

## NFA Dangerous Substances Classification

Class 2.8 (Containing 1-7) - 100 kg

## Control Parameters

| Component      | CAS No    | Korea                       | ACGIH - Biological Exposure Indices                                                      |
|----------------|-----------|-----------------------------|------------------------------------------------------------------------------------------|
| Cobalt, powder | 7440-48-4 | TWA: 0.02 mg/m <sup>3</sup> | 15 µg/L<br>Medium: urine<br>Time: end of shift at end of workweek<br>Determinant: Cobalt |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component      | CAS No    | Specifically Regulated<br>Chemicals | Highly Hazardous Chemicals |
|----------------|-----------|-------------------------------------|----------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                      | Not applicable             |

## CERCLA

Not applicable

| Component      | CAS No    | CERCLA Extremely<br>Hazardous Substances<br>RQs | Hazardous Substances<br>RQs | SARA 313 - Threshold<br>Values % |
|----------------|-----------|-------------------------------------------------|-----------------------------|----------------------------------|
| Cobalt, powder | 7440-48-4 | Not applicable                                  | Not applicable              | 0.1                              |

## CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H228 - Flammable solid. H302 - Harmful if swallowed. H330 - Fatal if inhaled. H317 - May cause an allergic skin reaction. H319 - Causes serious eye irritation. H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled. H341 - Suspected of causing genetic defects. H350i - May cause cancer by inhalation. H360F - May damage fertility. H410 - Very toxic to aquatic life with long lasting effects.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction. P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting. P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. P310 - Immediately call a POISON CENTER or doctor/physician. P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P284 - Wear respiratory protection.

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

# SAFETY DATA SHEET

Cobalt powder (Irregular spherical shape)

Revision Date 05-Jan-2024

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

## Prepared By

Health, Safety and Environmental Department

## Creation Date

07-Mar-2018

## Revision Date

05-Jan-2024

## Revision Number

1

## Revision Summary

Initial Release.

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

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**End of Safety Data Sheet**